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Austin Crime Analysis Comparing 2004 to 2009

The years 2004 and 2009

- National boom to bust
- Population Explosion in Texas and Austin
- Population Increase negatives
  - Crime Increase

Austin Facts

- 2004 Population – 692,102
- 2009 Population – 774,037
- Percent Increase – 11.8% Increase

- 2004 Combined Crime – 135,020
- 2009 Combined Crime – 160,424
- Percent Change – 18.8% Increase

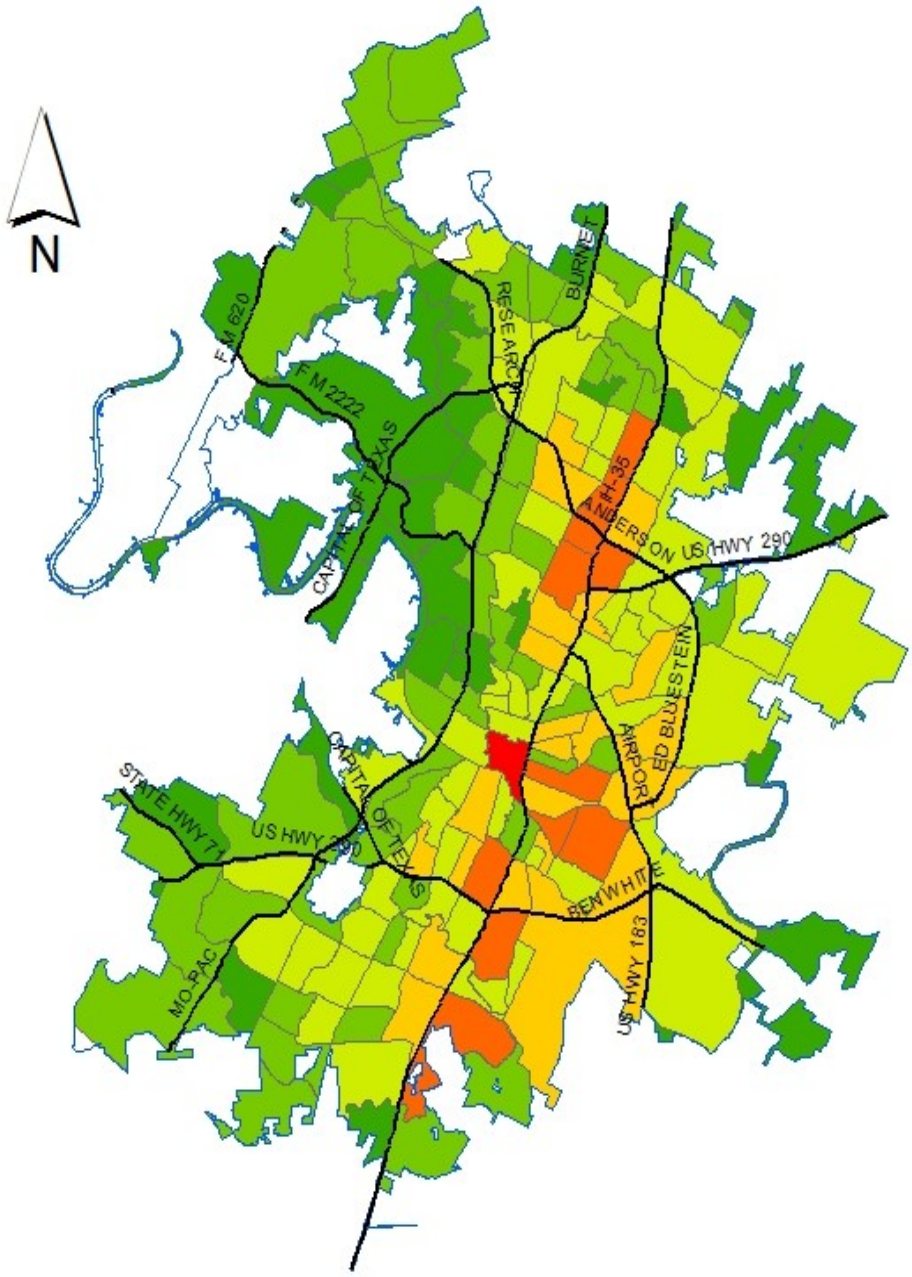
Problem with just facts and the spatial solution

- These facts don’t answer:
  - Where is crime occurring the most in Austin?
  - What areas have increased or decreased in crime from 2004 to 2009?
- Solution:
  - Spatially analyze the crime density in Austin and find the hot spots where crime has changed between 2004 and 2009.

Crime – what are we counting?

- “Data provided represents only calls for police service where a formal, written report was made and does not include other calls for police service that did not result in a written report.” – Austin Police Web Site
- Some Examples
  - murder
  - rape
  - robbery
  - aggravated assault
  - burglary
  - theft
  - motor vehicle theft
  - Arson

Austin Crime Per Census Tract 2004

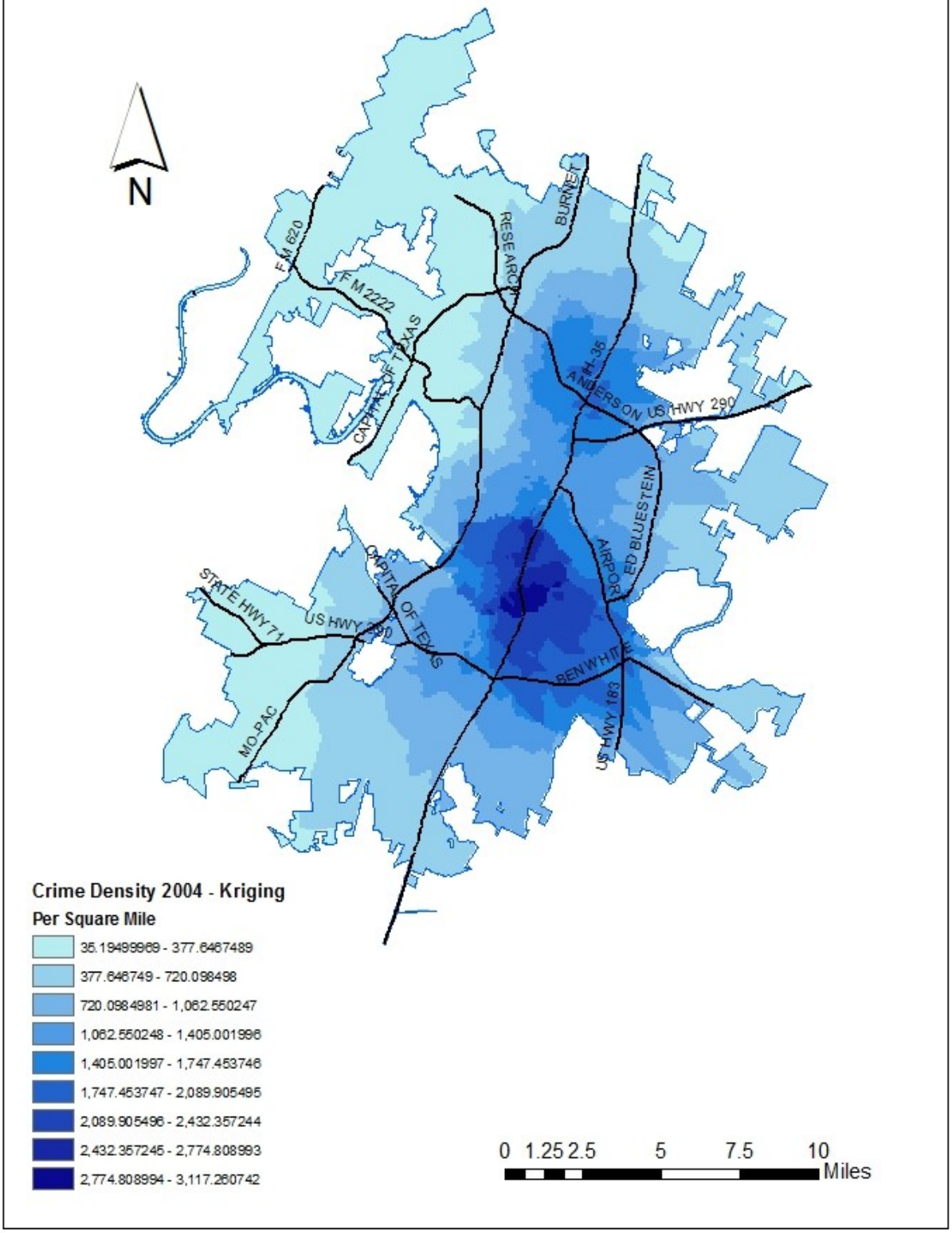


Crime Per Census Tract 2004

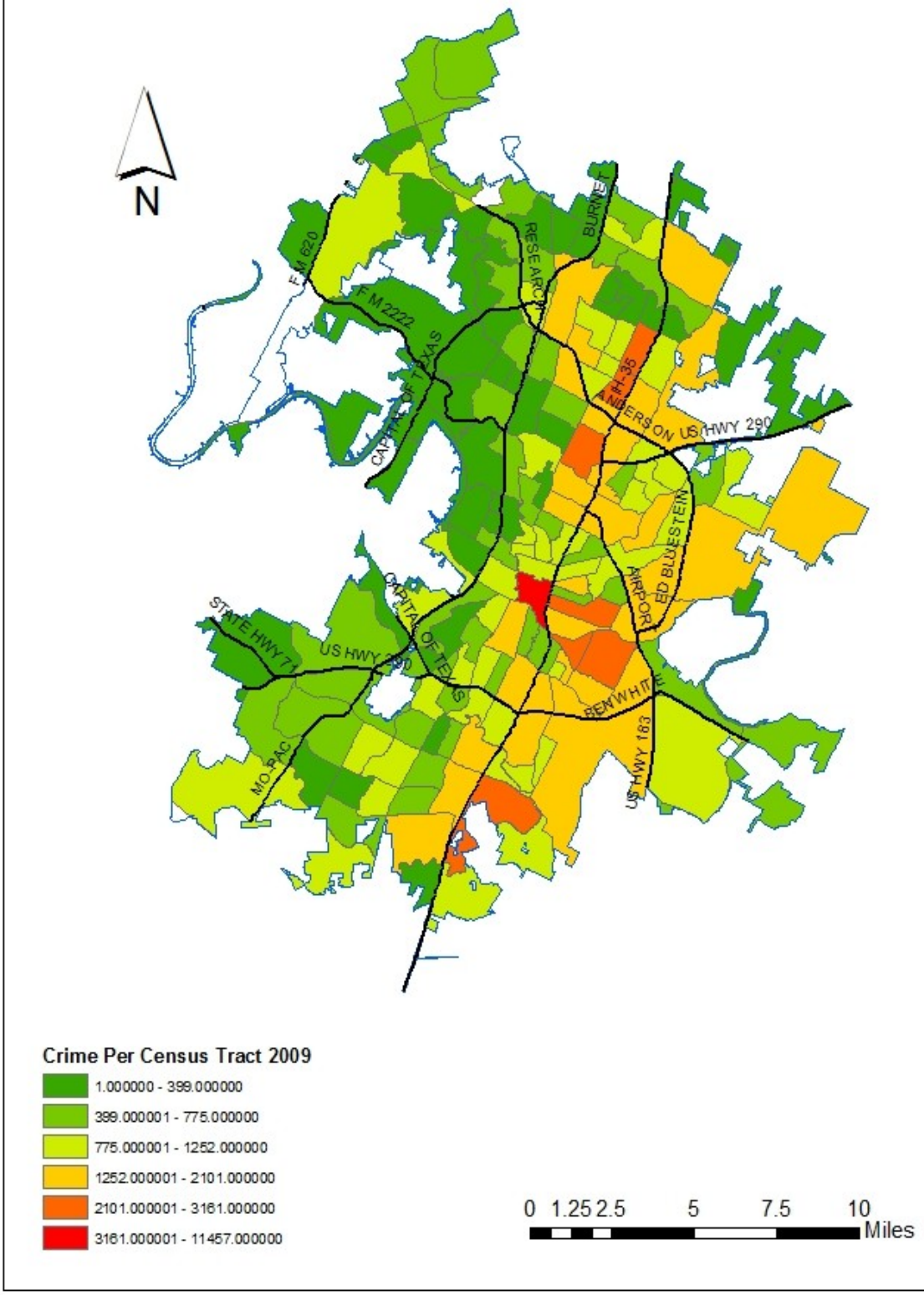
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0 1.25 2.5 5 7.5 10 Miles

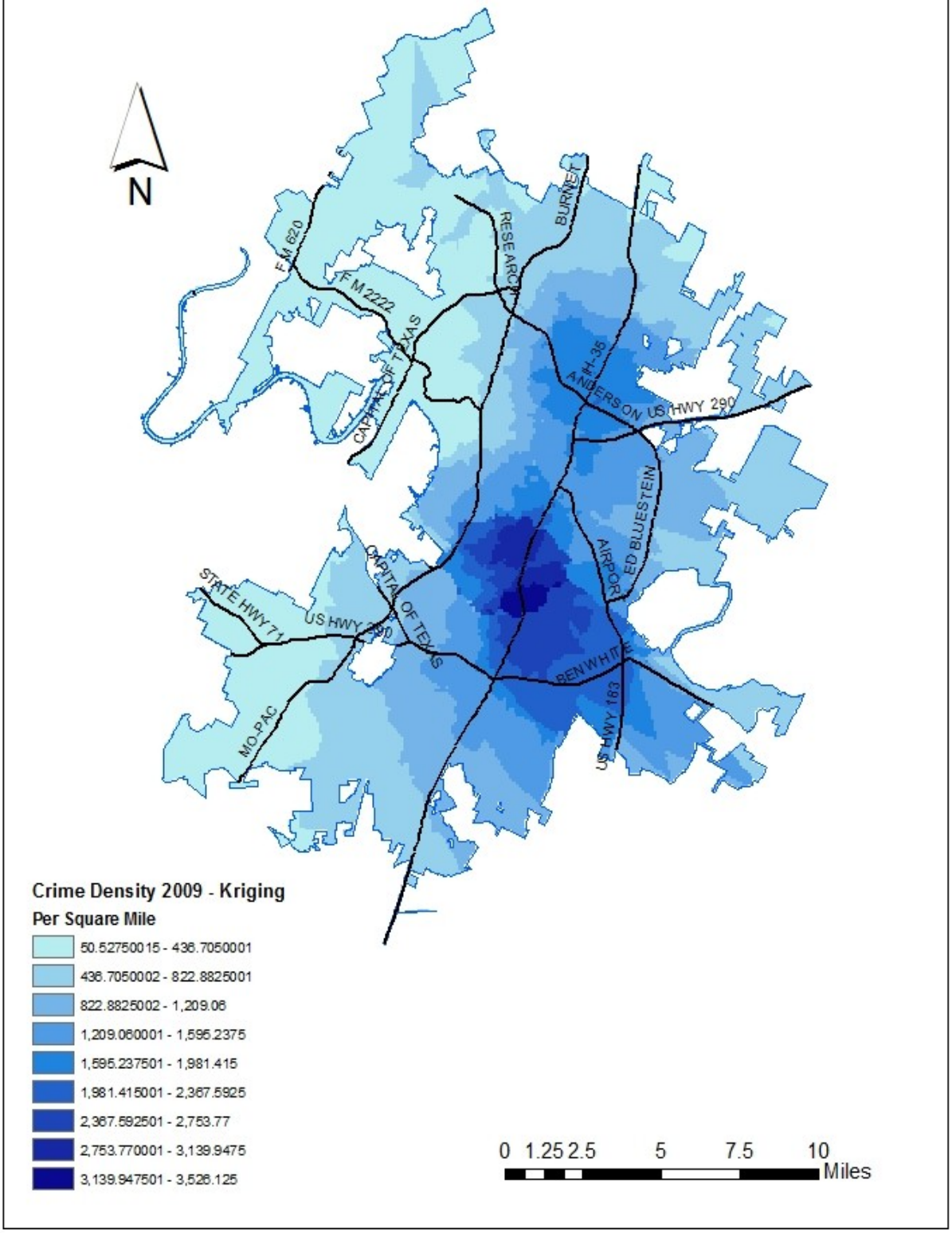
Austin Crime Density 2004



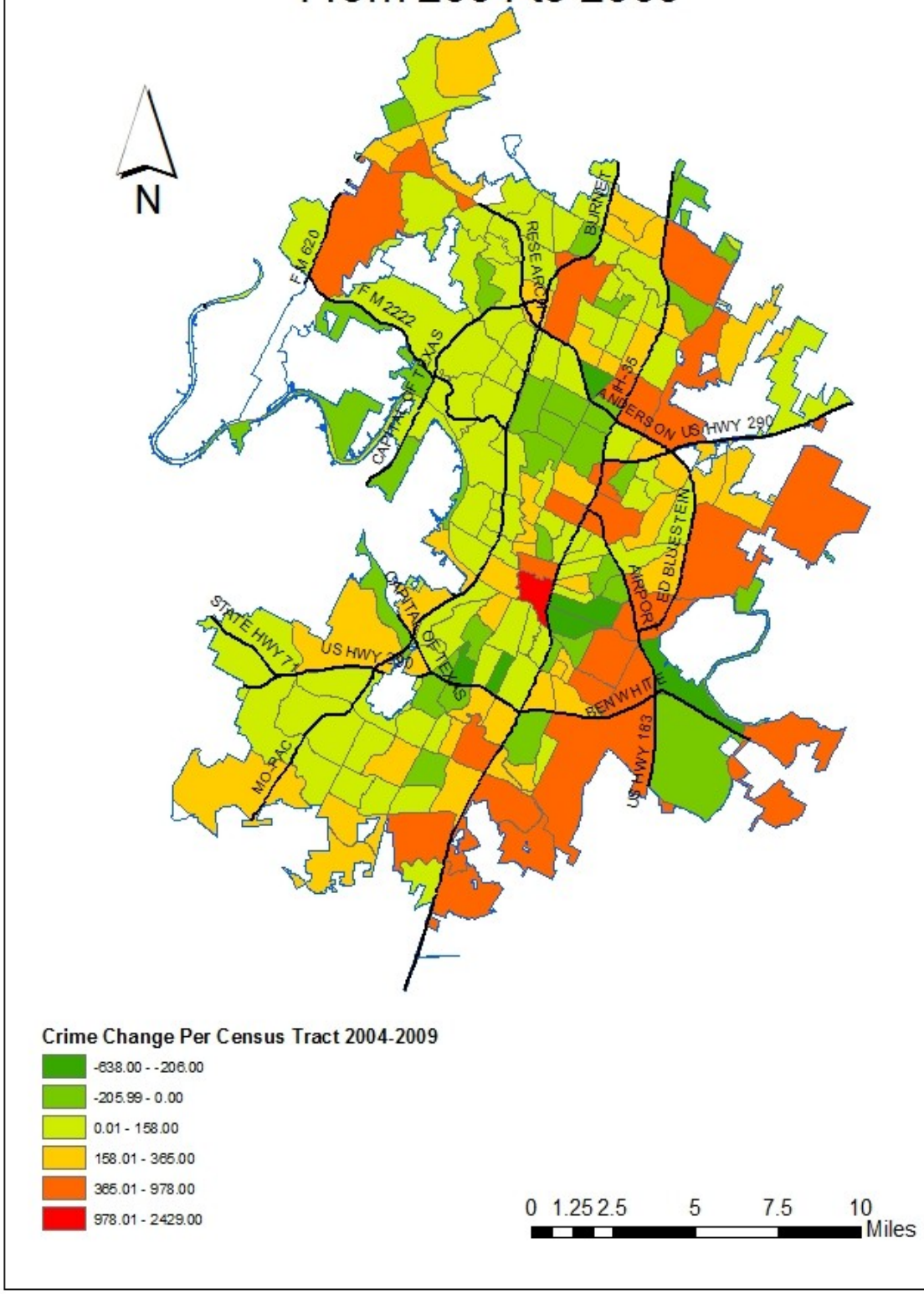
Austin Crime Per Census Tract 2009



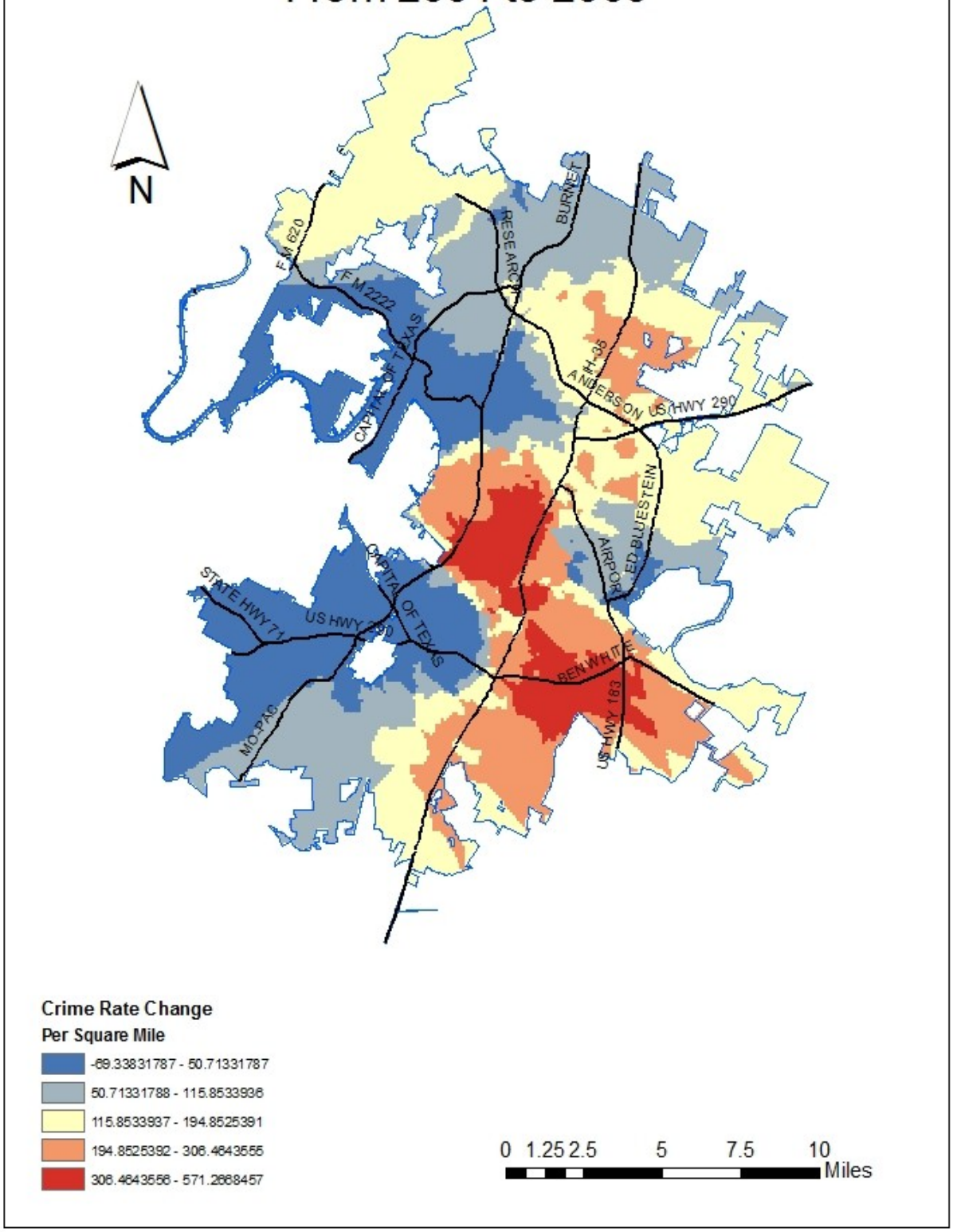
Austin Crime Density 2009



Change in Austin Crime Per Census Tract From 2004 to 2009



Change in Austin Crime Rate From 2004 to 2009



Conclusion

- The change in crime is clearly not evenly distributed in Austin
- Central, north central and southeast Austin have the highest increases in crime rates
- This information can be used by city planners and law enforcement

Steps in Implementing the GIS Model

- Gathered crime data from the City of Austin Website the data had total crime with census tract
- Census tract maps were gathered from the US census Bureau. The 2000 census tract map was used in this analysis
- Crime data was then converted to a database file. The crime by census tract data table was then joined with the 2000 census tract

Steps in Implementing the GIS Model (Cont.)

- The joined layer was then classified into graduated colors to visualize total crime per census tract for both the 2004 and 2009 data sets.
- Before any interpolation analysis was done, the centroids for the all the census block groups were found.

Steps in Implementing the GIS Model (Cont.)

- Interpolation was then done on these maps using kriging with a circular semivariogram model and a search radius type as variable and number of points 12. This produced two crime density maps for both years.
- The difference between the two kernal density maps was then performed with the raster calculator creating the hot spot map showing the increase in crime rate.

References

Web Site	URL
US Census Bureau	http://www.census.gov/geo/www/cob/tr2000.html
Capital Area Council of Governments	http://www.capcog.org/information-clearinghouse/
City of Austin Police Department	http://www.ci.austin.tx.us/police/censustract/censustract.cfm
Austin City Connection	http://www.ci.austin.tx.us/demographics/